EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 18

Basic Maths and Calculus

Area between Curves, Average Value and Revision

By - Excellence Programme

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Topics to be covered

01 5-minute substitution recap

02 Area between two curves

03 Which curve is above?

04 Average value of a function

05 Physics average velocity / mean value

06 Mixed indefinite + definite revision

07 Lock lines for the calculus block

Substitution from last class

Choose u, write du, rewrite the whole integral

Definite: change both limits with u

$\int_a^b f = F(b) - F(a)$ (no +C)

Check by differentiating

Quick check: substitution

Find $\int 2x \cos(x^2)\, dx$.

Solution: Quick check: substitution

Step 1: $u = x^2 \rightarrow du = 2x\,dx$

$\sin(x^2) + C$.

Rough work / extra work

Area between two curves

When one curve lies above another from x = a to x = b, the area between them is the difference of integrals.

$$\text{Area} = \int_a^b [y_{\text{upper}} - y_{\text{lower}}]\, dx$$

Always upper minus lower. Do not subtract in the wrong order.

Three board steps

1. Sketch both curves and mark a and b

2. Decide which is upper on [a, b]

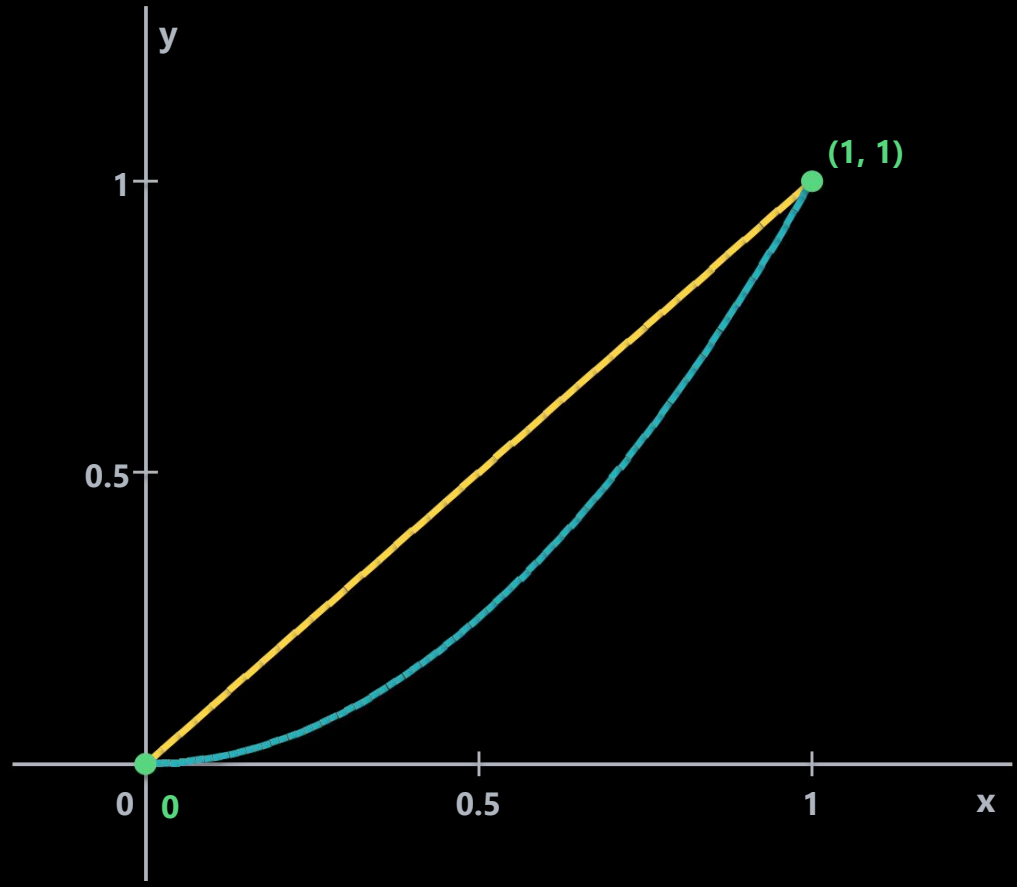
3. Integrate (upper − lower) from a to b

If they cross, split the integral at the intersection.

Region between $y = x$ and $y = x^2$

On [0, 1], the line $y = x$ is above the parabola $y = x^2$.

Area = $\int_0^1 (x - x^2)\,dx$



Area between line and parabola

Find the area between $y = x$ and $y = x^2$ from $x = 0$ to $x = 1$.

Solution: Area between line and parabola

Step 1: Upper = x, lower = x^2

Step 2: Area = $\int_0^1 (x - x^2)\,dx = [x^2/2 - x^3/3]_0^1$

Step 3: $= 1/2 - 1/3 = 1/6$

Area = 1/6 square units.

Rough work / extra work

Area between line and axis

Find the area between $y = 2x$ and the x-axis from $x = 1$ to $x = 3$.

Solution: Area between line and axis

Step 1: This is still ∫_1Â³ 2x dx (upper = 2x, lower = 0)

Step 2: = [x²]_1Â³ = 9 − 1 = 8

Area = 8 square units.

Area between two lines

Find the area between $y = 4$ and $y = x$ from $x = 1$ to $x = 3$.

Solution: Area between two lines

Step 1: On $[1, 3]$, $y = 4$ is above $y = x$

Step 2: Area $= \int_1^3 (4 - x)\,dx = [4x - x^2/2]_1^3$

Step 3: $= (12 - 9/2) - (4 - 1/2) = 15/2 - 7/2 = 4$

Area = 4 square units.

Find intersection first

Find the area between $y = x^2$ and $y = 4x - x^2$.

Solution: Find intersection first

Step 1: Intersection: $x^2 = 4x - x^2 \rightarrow 2x^2 - 4x = 0 \rightarrow 2x(x - 2) = 0$

Step 2: $x = 0$ or $x = 2$. On $[0, 2]$, upper $= 4x - x^2$

Step 3: Area $= \int_0^2 [(4x - x^2) - x^2]\, dx = \int_0^2 (4x - 2x^2)\, dx$

Step 4: $= [2x^2 - (2/3)x^3]_0^2 = 8 - 16/3 = 8/3$

Area = 8/3 square units.

Rough work / extra work

Average value of a function

The average height of y = f(x) on [a, b] is the area under the curve divided by the width.

$$f_{avg} = (1/(b - a)) \int_a^b f(x)\, dx$$

This is the mean value of f on that interval.

Do not forget to divide by (b − a).

Why this formula?

Area under f equals the area of a rectangle of width (b − a) and height f_avg.

$$\int_a^b f(x)\,dx = f_{avg} \cdot (b - a)$$

Solve for f_avg to get the formula.

Average of a line

Find the average value of f(x) = 2x on [0, 4].

Solution: Average of a line

Step 1: $f_{avg} = (1/4) \int_0â①´ 2x\,dx = (1/4)[x^2]_0â①´$

Step 2: $= (1/4)(16) = 4$

Average value = 4.

Rough work / extra work

Average of a parabola

Find the average value of $f(x) = x^2$ on $[0, 3]$.

Solution: Average of a parabola

Step 1: $f_{avg} = (1/3) \int_{0Â}^{3} x^2 \, dx = (1/3)[x^3/3]_{0Â}^{3}$

Step 2: $= (1/3)(9) = 3$

Average value = 3.

Average of cosine

Find the average value of $f(x) = \cos x$ on $[0, \pi/2]$.

Solution: Average of cosine

Step 1: Width = $\pi/2 - 0 = \pi/2$

Step 2: $f_{avg} = (2/\pi) \int_0^{\pi/2} \cos x \, dx = (2/\pi)[\sin x]_0^{\pi/2}$

Step 3: $= (2/\pi)(1) = 2/\pi$

Average value = $2/\pi$.

Physics: average velocity

If v(t) is given, the average velocity over $[t_1, t_2]$ is the mean value of v.

$$v_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} v(t)\,dt = \frac{\Delta x}{\Delta t}$$

Same formula as average value — now with time as the variable.

Rough work / extra work

Average velocity from v(t)

If $v = 2t$ (m s^{-1}) from $t = 0$ to $t = 4$ s, find average velocity.

Solution: Average velocity from v(t)

Step 1: $v_{avg} = (1/4)\int_0^4 2t\,dt = (1/4)[t^2]_0^4 = 4$

Step 2: Check: $\Delta x = 16$ m, $\Delta t = 4$ s → $v_{avg} = 4$ m s^{-1}

$v_{avg} = 4$ m s^{-1}.

Average of constant acceleration speed-up

If $v = 3 + 2t$ from $t = 0$ to $t = 3$, find v_{avg}.

Solution: Average of constant acceleration speed-up

Step 1: $v_{avg} = (1/3) \int_0^3 (3 + 2t)\, dt = (1/3)[3t + t^2]_0^3$

Step 2: $= (1/3)(9 + 9) = 6$

$v_{avg} = 6\ m\ s^{-1}$.

Mean of a linear a

If a = 4t from t = 1 to t = 3, find average acceleration.

Solution: Mean of a linear a

Step 1: $a_{avg} = (1/2) \int_{1Â}^{3} 4t\, dt = (1/2)[2t^{2}]_{1Â}^{3}$

Step 2: $= (1/2)(18 - 2) = 8$

$a_{avg} = 8\ m\ s^{-2}$.

Rough work / extra work

Indefinite integral lock lines

$\int x^{n}\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int \sin x\,dx = -\cos x + C \quad \cdot \quad \int \cos x\,dx = \sin x + C$

$\int e^{x}\,dx = e^{x} + C \quad \cdot \quad \int \frac{dx}{x} = \ln|x| + C$

Substitution undoes the chain rule

Revision: indefinite

Find $\int (4x^3 - 1/x)\,dx$ for $x > 0$.

Solution: Revision: indefinite

Step 1: $\int 4x^3\,dx = x^4$

Step 2: $\int -1/x\,dx = -\ln x$

$x^4 - \ln x + C$.

Revision: definite

Evaluate $\int_0^{\pi/4} \cos(2x) \cdot 2 \, dx$.

Solution: Revision: definite

Step 1: $u = 2x \rightarrow du = 2\,dx$; limits $0 \rightarrow \pi/2$

Step 2: $\int_0^{\pi/2} \cos u\,du = 1$

1.

Rough work / extra work

Revision: area between

Find the area between $y = \sqrt{x}$ and $y = x$ from $x = 0$ to $x = 1$.

Solution: Revision: area between

Step 1: On [0, 1], √x ≥ x

Step 2: Area = ∫_0^1 (√x − x) dx = [(2/3)xÂ³á➄ŸÂ² − x²/2]_0^1

Step 3: = 2/3 − 1/2 = 1/6

Area = 1/6 square units.

Revision: average value

Find the average value of $f(x) = 3x^2$ on $[1, 2]$.

Solution: Revision: average value

Step 1: $f_{avg} = (1/1) \int_{1Â^2} 3x^2\,dx = [x^3]_{1Â^2} = 8 - 1$

Average value = 7.

Rough work / extra work

Homework: between curves

Find the area between $y = 6 - x$ and $y = x$ from $x = 1$ to $x = 2$.

Solution: Homework: between curves

Step 1: Upper = 6 − x on this interval

Step 2: Area = $\int_{1}^{2} (6 - x - x)\,dx = \int_{1}^{2} (6 - 2x)\,dx$

Step 3: $= [6x - x^2]_{1}^{2} = (12 - 4) - (6 - 1) = 3$

Area = 3 square units.

Homework: average velocity

If $v = t^2$ from $t = 0$ to $t = 3$, find v_{avg}.

Solution: Homework: average velocity

Step 1: $v_{avg} = (1/3) \int_{0Â}^{3} t^2 \, dt = (1/3)[t^3/3]_{0Â}^{3} = 3$

$v_{avg} = 3 \text{ m s}^{-1}$.

Calculus block lock lines

Area between curves: ∫ (upper − lower) dx

Average value: f_avg = (1/(b−a)) ∫_a^b f dx

v_avg = (1/Δt) ∫ v dt = Δx/Δt

Differentiation ↔ Integration are reverse operations

Rough work / extra work

Practice

1. Sketch every area-between-curves question before integrating.

2. Memorise the average-value formula with (b − a) in the denominator.

3. Redo one substitution + one definite + one average question.

4. Write a one-page summary of the whole calculus block.

Next chapter after this block: Vectors.